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ChatGPT in thematic analysis: Can AI become a research assistant in qualitative research?

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Abstract
Despite an emerging body of scholarship on applying generative AI (GenAI) to qualitative data analysis, this area remains underdeveloped. This article evaluates how GenAI can support thematic analysis using a publicly available interview dataset from Lumivero. It introduces Guided AI Thematic Analysis (GAITA), an adaptation of King et al.'s (2018) Template Analysis. This framework positions researchers as a reflexive instrument and intellectual leader while thoroughly guiding GPT-4 in four stages: data familiarization; preliminary coding; template formation and finalization; and theme development. Additionally, the article proposes the ACTOR framework, a simple approach to combining different effective prompting techniques when working with GenAI for qualitative research purposes. Findings reveal GenAI's capacity for analyzing the data, generating codes, subcodes, clusters, and themes, along with its adaptive learning and interactive assistance in organizing unstructured data and developing trustworthiness. However, this model has some key limitations in terms of its restricted context window for processing large datasets, its inconsistent outputs requiring multiple prompt attempts, the need to move across workspaces, and the lack of relevant training data for qualitative research purposes.

Keywords ChatGPT · Coding · Categorizing · Generative AI · Qualitative data analysis · Thematic analysis · Template analysis · Reflexivity

1 Introduction

In 1984, sociologist Gerson (1984) proposed that artificial intelligence (AI) could act beyond a clerical capacity as a research assistant in qualitative research with proper 'sociological knowledge' of the research topics. His hypothesis might now be achievable through recent technological advancements, following OpenAI's release of ChatGPT to the public in November 2022. Since then, there has been extensive debate about the potential of generative AI (GenAI) to transform the way we conduct research (Sallam 2023). It has been suggested that GenAI can be applied in various professional areas, including climate

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change-related initiatives (Vagheti et al. 2024; Biswas 2023), health care (Li et al. 2024), education (Firat 2023), and evidence review (Nguyen-Trung et al. 2024). Qualitative research is not excluded from this movement. As of this writing, the field has witnessed the incorporation of GenAI into the top three computer-assisted qualitative data analysis software programs (CAQDAS), namely MaxQDA, Atlas.ti, and NVivo. Moreover, several recently developed software programs have emerged and labeled themselves as qualitative AI (or qualAI for short), such as QualAI (by QUALAI) and CoLoop (by CoLoop). Yet, concerns exist about GenAI producing inconsistent results, with tendencies to hallucinate or confabulate responses (Haman and Školník; 2023 van Dis et al. 2023). So, can GenAI effectively serve as a research assistant in qualitative data analysis?

In this article, I aim to answer this question through using ChatGPT to carry out a thematic analysis of qualitative data. The article is organized as follows: Section 2 provides a brief overview of the role of technologies, machine learning (ML) and AI in social science research and qualitative data analysis (QDA). Section 3 proposes a Guided AI Thematic Analysis (GAITA) approach and describes the steps and prompting techniques (i.e., the ACTOR framework) in an adapted version of Template Analysis proposed by King et al. (2018). Section 4 presents the results of this process in four steps. The study used a sample qualitative research project, "Environmental Change in Down East" (Campbell and Meletis 2011; Lumivero 2023), with Lumivero's permission.

2 AI and technologies in qualitative research

2.1 Early use of AI in qualitative data analysis

The idea of AI has existed since the 1950s, perhaps gaining attention with the publication of Turing's (1950) paper 'Computing machinery and intelligence' which posed the classic question, "Can machines think?". With the prospect of machines being capable of competing with humans in intellectual fields (Turing 1950), this technology presented the possibility of automating research tasks.

In qualitative research, the agenda for using technologies to empower qualitative research practice began in the 1980s when qualitative inquiry took a technological turn in QDA (Seidel and Clark 1984). The 1980s and subsequent decades witnessed a rise of many Computer-Assisted Qualitative Data Analysis Software (CAQDAS): NUS*IST (NVivo's predecessor) (1981), followed by the Ethnograph (1982), MaxQDA (1989), HyperRESEARCH and Atlas.ti (1991), Transana (2001), QDA Miner (2004), Ethnonotes (later became Dedoose in 2009) (2003), Discovertext (2009), Quirkos (2014), and Taguette (2019) (Fassnacht and Woods 2005; Spiers 2004; Hesse-Biber et al. 1991; Jessica 2019; Kuckartz and Kuckartz 2002; LaPan 2013; Lewis and Maas 2007; Lieber et al. 2003, 2021; Rampin and Rampin 2021; Richards and Richards 1991; Richards 2002; Seidel and Clark 1984; Turner 2014). Since 2008, CAQDAS programs, such as NVivo and Atlas.ti, have had the capacity to support not just text-based analysis but also multimodal data analysis—including audio, photo, video, multimedia, social-media, geodata, and survey datasets—and are primarily used for data management and analysis (Woods, Paulus, et al. 2016, p. 15).

In parallel, experiments with using AI in QDA took advantage of the development of expert systems since 1969—systems designed specifically to mimic human decision-making processes (Russell and Norvig 2010). These 'expert systems', also called 'knowledge systems', were believed to have the potential to 'understand' natural language and serve as "supplements or sometimes replacements for complex decision support in professional settings, such as medical diagnostics" (Elish and Boyd 2018, p. 61).

In this context, Gerson (1984) proposed a hypothetical system called "Automatic Notebook for Social Research" (ANSR), which was believed to be able to act beyond a clerical capacity as a research assistant who could "take over portions of the most routine and highly defined tasks" (Gerson 1984, p. 71). To fulfill this role, the proposed system would be built on AI technology and be capable of understanding "sociological knowledge" (e.g., family systems). However, that hypothesis was not achievable due to the high cost of technologies, the limited capacity of personal computers, and the limited capacity of expert systems, especially in learning from a large dataset. Following Gerson's hypothetical system, Brent and Slusarz (2003) introduced Qualrus, released in 2002, as the first CAQDAS package using "intelligent computational strategies," including case-based reasoning, natural-language understanding, and intelligent agents. Also based on expert systems, the Qualrus system was limited to "production rules" (i.e., "If X then Y") (Brent and Slusarz 2003, p. 287).

2.2 Machine learning in qualitative data analysis

In the last two decades, the field of QDA has witnessed the integration of machine learning (ML) into CAQDAS such as DiscoverText, QDA Miner, and Leximancer (Paulus 2023). DiscoverText, a web-based text analytics software launched by Texifter, LLC in 2009, enhances the accuracy and efficiency of classifying text data (especially Twitter data) by combining human and machine capabilities in a continuous iterative process between human input and ML (Shulman 2011; Silver 2019). While Leximancer automatically generates concept seeds from text based on the co-occurrence and correlation of words, which then inform the creation of more complex concepts through a thesaurus-like feature that refines these concepts further based on the data, QDA Miner offers tools like Cluster Extraction and Query by example, which assist in coding by suggesting similar text segments and support user-driven refinement, allowing the software to adapt its suggestions based on user interactions (Lewins and Silver 2020; Silver and Lewins 2020).

However, these applications of ML were primarily for certain text analysis functionalities, including automatic coding, concept generation, and clustering, with limited capacity to generalize beyond *predefined algorithms* and tasks, and more importantly, to engage in *meaningful conversation* with human researchers. In the broader context, current ML models designed for QDA purposes such as Cody and Aeonium (Rietz and Maedche 2021; Drouhard et al. 2017; Hong et al. 2022), focus largely on qualitative coding, with key features such as suggesting codes and highlighting keywords based on human inputs and rules. However, these features, such as word counts, keywords, and semantic analysis, tend to decontextualize the words or instances and do not represent the concepts that human researchers understand and use in their research (Chen et al. 2018). In QDA, the meaning and significance of words or instances are not often determined by their frequencies or correlations with others but by the shared patterns among codes, categories, and cases and their connection with the contexts in which participants live and work (Saldaña 2016; Miles and Huberman 1994). Therefore, in addressing the hypothetical vision set by Gerson (1984) about the prospect of AI performing beyond a clerical capacity, I argue that the current applications of ML in qualitative research are still somewhat limited in offering opportunities for flexible and deeper engagement in response to emerging issues and new tasks during the analysis process. Furthermore, there remains a hesitance among qualitative researchers to use ML and AI in QDA due to their complexity (Chen et al. 2018). AI-assisted systems should be simple and accessible for non-technical users.

2.3 The emergence of generative AI in QDA

Different from previous AI technologies, Generative AI (GenAI) has taken advantage of the latest developments including big data, natural language processing, neural networks, computer vision, and deep learning. With these developments, AI can now process an extensive dataset of various types, including not just structured but unstructured data, in an accelerated timeframe (Elish and Boyd 2018). A GenAI model like GPT (*Generative Pre-trained Transformer*) can be trained and learn with a "large amount of publicly available digital content data" (natural language processing) and then respond in human-like conversation in multiple languages (Baidoo-Anu and Owusu Ansah 2023, p. 53; Lund and Wang 2023). This implies that non-technical users can easily engage with AI chatbots in QDA.

GenAI can be applied in QDA in three scenarios: the integration of AI into existing CAQDAS packages; the development of new AI qualitative research applications; and the use of chatbot tools (Silver 2023) (see Appendix 6). The integration of GenAI in current CAQDAS packages such as MaxQDA (MaxQDA 2023), Atlas.ti (ATLAS.ti 2023) and NVivo (Lumivero 2023) comes with the promise of "AI can automatically unveil what is there" (Paulus and Marone 2024, p. 4), including the capacities to suggest codes and themes, summarize, produce memos, and 'chat with data'. These packages can offer a single analytic workspace, leverage existing systems' features, and address data security concerns through the assurances provided by the companies that develop and commercialize them. The second scenario comes with considerable diversity dependent on each company. Though built differently, current new qualitative AI software such as Canvs, CoLoop, and QInsight offer key capacities including dealing with multimodal data, producing summaries, chatting with data, and developing analytical matrices (Canvs 2024; Friese 2023; CoLoop 2024; Q-insight 2024). One disadvantage of this second scenario is the requirement of learning a new system and its pricing.[1]

The third scenario involves the use of AI chatbots such as OpenAI's ChatGPT, Google's Gemini, Microsoft's Copilot, Perplexity AI's Perplexity, and Anthropic's Claude in supporting data analysis. While most of these chatbots are free, the advanced use of their models with enhanced data security options requires a paid subscription. With this paid option, the use of AI Chatbots might be more cost-effective than the aforementioned scenarios, as users can flexibly choose the monthly subscription (about $22) and can cancel anytime.

2.4 Current applications of GenAI in QDA

The above events signify a significant turning point that gives rise to an emerging scholarship on applying GenAI in QDA (Hitch 2023; De Paoli 2024). GenAI has been applied in various analysis approaches such as context analysis (Chew et al. 2023), discourse analysis (Fan et al. 2023; Curry et al. 2024), and grounded theory (Sinha et al. 2024; Wachinger et al. 2024). Thematic analysis with its relative flexibility in adopting a suitable philosophical position, is the most widely used approach in experiments

[1] I received a monthly quote for an independent plan from CoLoop in May 2024 at $200 (plus VAT) that offers 12 projects/year with 600 h transcription or translation and 600 Analysis Grid Questions.

with GenAI (Morgan 2023; Hitch 2023; Hamilton et al. 2023). These experiments show that various GenAI tools [e.g., ChatGPT, Google's Gemini (previously, Bard)] are capable of supporting qualitative researchers in various tasks, especially summarizing large amounts of text, suggesting codes based on text segments, and generating themes (Drápal et al. 2023; De Paoli 2024; Prescott et al. 2024).

However, a review of the current literature suggests some areas that need further research. First, most existing experiments (De Paoli 2023, 2023; Lee et al. 2024; Prescott et al. 2024) used the thematic analysis approach proposed by Braun and Clarke (2006), which has since evolved through new advancements in their recent work (Braun and Clarke 2019, 2022). There is a need to experiment with GenAI using alternative thematic analysis approaches. Template Analysis (King et al. 2018) offers a balanced approach that can flexibly integrate the strengths of the Big Q thematic analysis (i.e., reflexive thematic analysis by Braun and Clarke (2019) and small q thematic analysis (i.e., Guest et al.'s (2012) coding reliability thematic analysis) (see the next section for further discussion).

Second, there is a need for a comprehensive end-to-end thematic analysis. Most studies focus only on spefic phases in Braun and Clarke's (2006) approach, such as Phase 2 (initial coding generation) and Phase 3 (initial development of themes) (De Paoli 2024; Drápal et al. 2023). Other researchers have focused on immediate generation of themes. For instance, Hamilton et al. (2023) compared human-generated themes and ChatGPT-generated themes based on 1125 human-chosen statements from interviews. They fed all the statements to ChatGPT with direct requests for theme generation. While their study's strengths lie in comparing human-led analysis and ChatGPT outputs, asking ChatGPT to jump immediately into generating themes while omitting the stages of coding, categorizing, recoding/recategorizing, case comparison, and reflexive supervision and journaling could risk oversimplifying data.

In another notable study, Morgan (2023) used ChatGPT to analyze two datasets he had already analyzed. His prompts were based on his previous understanding of the data, which helped him better judge the themes generated by ChatGPT. His use of ChatGPT tended to be "query-based," focusing on asking this AI chatbot to generate an understanding of certain topics (e.g., "Give me a long description about the topic of keeping balance between school and social life") without applying the full six phases of reflexive thematic analysis by Braun and Clarke (2006). Notably, most of ChatGPT-generated themes tended to be topic summaries (i.e., topic-based nouns) rather than themes as *patterns of shared meanings* across the dataset (Braun and Clarke 2022). The query-based approach could serve as an alternative to the dominant coding approach in QDA. However, with the present understanding of ChatGPT and GenAI technologies, this approach needs further development and the establishment of a systematic procedure for reflexive monitoring, evaluation, self-reflection, and learning.

Third, there is a lack of integration of AI into human researchers' analysis. Current research tends to compare between human-generated and GenAI-generated codes/themes. This testing approach helps validate if GenAI's models can produce or predict themes that are as good as those from human researchers. For instance, using ChatGPT in grounded theory and framework analysis approaches, Wachinger et al (2024, p. 6) observe that "ChatGPT was able to produce thematic insights that to a considerable degree aligned with or resembled those produced by an experienced human researcher." Similarly, De Paoli (2024) used LLMs Mistral-7b-Instruct-V0.22 and Llama-8b-Instruct3 to conduct an evaluation of semantic similarities between human themes and GPT3.5-generated themes. He found that "the LLM produces themes which, at least in part, are semantically alike to themes produced by human analysts." (De Paoli 2024, p. 28).

However, AI's capacity to produce similar themes is less meaningful in practice if researchers do not consider using AI-generated outputs to advance their analysis. Even if GenAI can generate themes comparable to human researchers, they cannot (and *should not*) perform the full data analysis process alone without the researchers' guidance and interpretations. Importantly, for qualitative research, researchers need to maintain their role as the primary decision-makers - not just over the dataset, but over the entire analytical process. Consequently, existing experiments comparing human- and AI-generated outputs often conclude by recommending that in further research, qualitative researchers need to apply "full agency" throughout the analysis process (Drápal et al. 2023, p. 9). This "full agency' implies the leading role of qualitative researchers in providing inputs, suggestions, judgements, and making decisions, with GenAI serving as their research assistant, not an autonomous agent. This approach ensures the alignment with qualitative research's principles that see researchers as the primary instrument (Braun and Clarke 2022). Therefore, there is a need to integrate GenAI as a research assistant into the full workflow of thematic analysis to see how researchers and GenAI interact and colaborate to produce results.

Finally, there is a need for simple AI use guides that are accessible to non-technical users. In their review of machine learning in assisting literature review, Khalil et al. (2022) found out that current applications such as LitSuggest, Rayyan, AbstracKr pose a challenge to non-technical users who lack expertise in computer science and algorithms. In this regard, GenAI-powered chatbots such as ChatGPT hold an advantage due to their ease of use (Nguyen-Trung et al. 2024). Yet, current experiments of AI in QDA have not yet offered user-friendly guidelines. For instance, instead of using web-based chatbots, the procedures proposed by Drápal et al (2023) and De Paoli (2024) involve the use of Application Program Interface (API) and require expertise in coding programs such as Python to support GenAI interaction. Additionally, there is a need to translate the most effective prompting techniques into simple frameworks that help non-technical users facilitate their approach (see for instance, Turobov et al. 2024).

To address the above gaps, this study employed GPT-4 model (subscription version) as a research assistant for thematic analysis. Rather than designing a traditional experiment approach comparing control and treatment groups, the study adopts a qualitative approach. Unlike the existing studies (De Paoli 2023; Drápal et al. 2023), I developed an user-friendly, adapted Template Analysis with AI guideline to facilitate a *Guided AI Thematic Analysis* (*GAITA*) approach for directing and supervising GPT-4. I offer detailed documentation of a four-step process with a simple prompting framework ACTOR. Throughout this process, my supervision, reflexive notes, judgement, and adjustment are key to ensuring the meaningful results of GPT-4's outputs (see Table 1). This article thus presents critical reflections on GPT-4's capacity to execute my guidance in Template Analysis.

3 Guided AI thematic analysis (GAITA)

The ethical concerns of using ChatGPT in scholarly research have been subject to debate. Some concerns are related to data ownership and rights (e.g., sharing data with commercial entities), data privacy and transparency (e.g., sharing individual data) (Davison et al. 2024). OpenAI provides an option in *Settings* that allows users to turn off "Improve the model for everyone", which means that OpenAI promises it will not use the chat data for

Table 1 Steps, prompts, AI's tasks and researcher's role in template analysis

Steps	Substeps	Prompts, GPT-4's Tasks	Researcher's role
Step 1: Data familiarization	Produce summaries of each transcript in different lengths	Prompt 1.1: produce a general summary of the first transcript Prompt 1.2: produce a highlight in three bullet points	Read these different summaries to have a quick understanding of the dataset. Read the original transcripts to check this understanding. Read back and forth between GPT-4-generated contents and original transcripts. Write initial impressions and reflexive notes.
	Produce a summary of the whole dataset	Prompt 1.3: produce a summary of all interviews	
Step 2: Preliminary coding	Code the first transcript	Prompt 2.1: code the first transcript to produce a list of initial codes	Develop a Google Sheet file to record codes by ChatGPT. Check the list of initial codes produced by ChatGPT. Edit the list by removing incorrect codes, renaming codes, or adding new codes, and subsequently produce a revised list of initial codes. Write code reflections.
	Code the remainder of transcripts	Prompt 2.2: study the revised list of initial codes, preparing for coding the remainder of the transcripts Prompt 2.3: apply the revised list of initial codes to code one transcript at a time in the remainder of transcripts	Check carefully the codes produced by ChatGPT after each round of coding. Write reflexive notes documenting decisions and changes made during coding.
	Produce a refined list of codes after coding the whole transcripts	N/A	Use the Google Sheets file to refine the list of codes produced by removing wrong codes, renaming codes, or adding codes. Produce the refined list, in which all the codes are named, defined, and linked to quotations. Write reflexive notes on the revised code list.

Table 1 (continued)

Steps	Substeps	Prompts, GPT-4's Tasks	Researcher's role
Step 3: Template formation and refinement	Group codes into clusters to produce an initial template	Prompt 3.1: group codes into clusters while maintaining the links with quotations	Use the Google Sheets file to record coding development. Check the logical links among clusters, codes and quotations. Check for any omitted codes or missing data. Revise the list of clusters and codes to form the initial template. Write reflexive notes on individual clusters and the relationships among their codes.
	Generate subcodes for each code in a single cluster	Prompt 3.2: develop subcodes for each code	Ensure the logical link between the cluster, codes, subcodes, and quotations. Ensure GPT-4 captures nuances from transcripts' quotations. Write reflexive notes on the relationships among codes, subcodes and the original transcripts.
	Revise coding for each code under each cluster	Prompt 3.3: check for overlapping codes	Compare the list of subcodes and quotations under each code and identifying the overlapping subcodes/quotations. Decide how to arrange the coding structure.
	Produce the final template	Repeat Prompts 3.2 and 3.3 for the remainder of the clusters	Check the logical links among clusters, codes, and quotations and revising if needed. Produce the final template. Write reflexive notes on the final template and coding structure.
Step 4: Theme development	Generate themes based on the final template	Prompt 4.1: generate list of themes	Review and revise the themes generated by linking the themes' names and descriptions to the context of the dataset. Write thematic notes, focusing on meaning and relevance of each theme.
	Generate an overarching theme to develop a narrative	Prompt 4.2: generate an overarching theme	Review and revise the central theme and the unifying narrative it presents, making sure they are meaningful. Write reflexive notes on thematic relationships and the overarching theme.

training its models. If researchers want to use ChatGPT or GenAI in QDA, it is critical to include the plan of how they will be using AI in their ethics application. For instance, in his successful ethics application for the use of ChatGPT in his interview study, Kaufman (2024) included the careful consideration of how GenAI would affect participants, data security and privacy. In his Explanatory Statement, he described in details the stages at which he would use GenAI, the risks of using GenAI and ChatGPT in research, and the plan for addressing these risks (e.g., anonymizing data before submitting it to ChatGPT). In his Consent Form, he included a request asking participants to allow the research team to use GenAI (ChatGPT in this case) in QDA.

As this study started early in 2023, when the use of ChatGPT and GenAI was rare and my university did not issue a complete guide on using AI in research, I chose the sample public project "Environmental Change in Down East" provided by Lumivero (2023) to avoid any ethical concerns. This sample project includes 11 short interviews conducted in 2008–2009 by scholars from Duke University's Nicholas School of the Environment at their Marine Lab in Beaufort, N.C, which aimed to examine how communities in the Down East region of Carteret County, North Carolina, USA, perceived environmental change (Campbell and Meletis 2011). Each interview transcript is about 6 pages long or about 2000 words. Since this dataset contains multiple topics such as connections to Down East, vision for the future of Down East, barriers to and opportunities for achieving your vision, I chose one research question to narrow down my study, that is, "How do these people connect to Down East?". As a result of this choice, the first two interview questions and the respondents' answers were chosen. The total word count for responses to these two questions across 11 interviews is 6306 words. I acknowledge that this selection could limit the scope of my application of ChatGPT in thematic analysis.

To facilitate QDA, I employed a thematic analysis approach. Thematic analysis can be classified into three broad types, ranging from small q qualitative research to Big Q qualitative research [see Kiddler and Fine (1987) for the origin of this classification]: coding reliability, codebook, and reflexive thematic analysis (Braun and Clarke 2022). Of these, coding reliability thematic analysis (small q qualitative) tends to apply positivist and/or realist assumptions (e.g., the use of intercoder agreement check to measure validity), while reflexive thematic analysis (Big Q qualitative) rests on non- (or anti-) positivist principles that consider knowledge production as "partial, situated, and contextual" with the researcher's subjectivity as a crucial resource, and the codebook approach lies somewhere in between (Braun and Clarke 2024). Because the reflexive thematic analysis by Braun and Clarke (2019) demands considerable researcher reflexivity and extended engagement with the data, I consider it less suitable for my current study. Meanwhile, as a qualitative researcher, I do not subscribe to the coding reliability approach, which advocates for the accuracy and objectivity of data analysis by employing multiple coders, requiring intercoder agreement check, and developing an early codebook or coding scheme. Hence, for this study, I adopted the Template Analysis approach proposed by King et al. (2018). Their approach occupies a middle ground between the above-mentioned approaches, incorporating principles from both Big Q and small q traditions in qualitative data analysis. For instance, while Template Analysis proposes the use of some a priori themes in the very early stage of data analysis (i.e., preliminary coding) based on prior projects, existing theories, or researchers' experience (King et al. 2018, p. 202), it remains open to the emergence of new themes during later stages of data analysis. King et al. (2018, p. 200) make this clear: "In any case, a priori themes in thematic analysis should always be considered as tentative. If they do not prove to 'fit' with the data they can be revised or deleted just like any other theme".

This approach relatively fits my *pragmatic* orientation, which relies on *abductive reasoning*- moving back and forth between induction and deduction (Morgan 2007, p. 71) - rather than committing exclusively to either only deductive or inductive reasoning.

Based on King et al.'s model (2018), I developed a Guided AI Thematic Analysis (GAITA) including four steps (see Fig. 1). Specifically designed to enhance the effectiveness when working with ChatGPT, the Guided AI Thematic Analysis (GAITA) simplifies King et al.'s (2018) original eight-step Template Analysis. As seen in Fig. 1, the first two steps are similar to the original version: data familiarization and preliminary coding. The differences are in Steps 3 and 4. Step 3 focuses on template development by combining the tasks of "Clustering", "Developing the initial template", "Modifying the template" and "Defining the 'final' template". In this step, I refrained from having GPT-4 early produce and define themes. Instead, Step 4 is specifically designed for theme generation. Different from the original Template Analysis approach, theme generation is separated from template formation and refinement for two reasons. First, this separation allowed me to train GPT-4 on the differences between clusters and themes. As shown in the results section below, GPT-4 demonstrated that it could not distinguish key concepts in QDA such as code, cluster/ category, template, and theme. This issue could be due to its lack of training on QDA-specific data. Thus, it is important to guide GPT-4 in the cluster and theme development process. Second, the separation helped prevent GPT-4 from jumping too early into theme development before a careful analysis of the dataset. Note that GPT-4 can generate themes immediately after we give it the transcripts. Yet, we could not check whether these early-generated themes are mature or meaningful. Moreover, the production of an overarching theme was also not present in King et al.'s (2018) approach but I believe this is important for generating a narrative that unifies themes, clusters, and codes. In this study, I omitted the steps of using the final template to interpret data and writing up, due to time constraints and resources. In

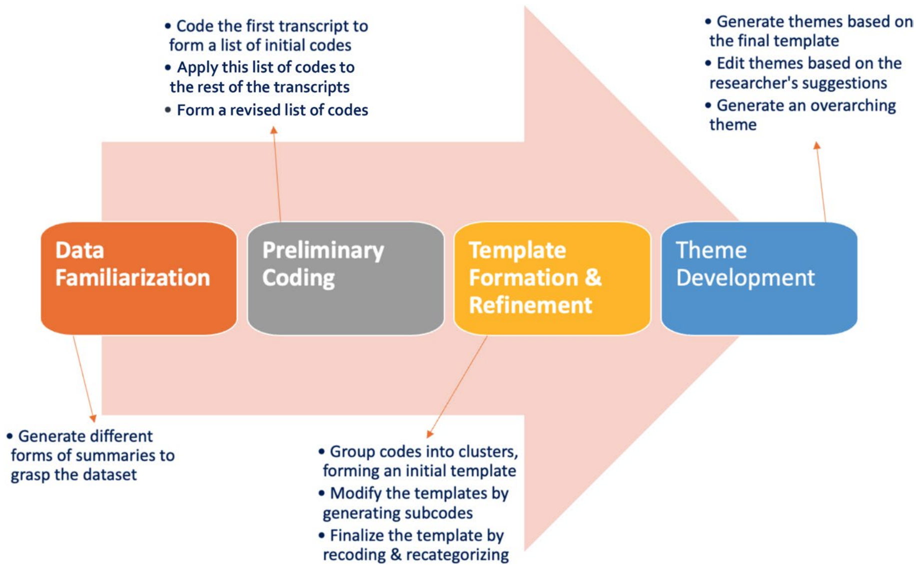

Fig. 1 A Guided AI Thematic Analysis procedure

practice, Step 4 - Theme development - can also include interpretation and writing up the findings.

Table 1 describes the main prompts (i.e., key commands) in correspondence with each step of the GAITA approach. The detailed prompts are provided in the results section. I numbered those prompts to help readers easily follow the steps I took. The researcher's role was to lead and monitor GTP-4 throughout the analysis process. My responsibilities included, but were not limited to, defining and refining my inputs (e.g., research needs, prompts), evaluating GPT-4's outputs, requesting GPT-4 to revise its outputs, and documenting my decisions along the analysis journey.

Please note that in Table 1, only the key tasks are presented for each prompt. The full prompts are provided in the result section. Based on current prompting guidelines (Lin 2024; Eager 2023) and inspired by Presseau et al.'s (2019) Action, actor, context, target, and time (AACTT) framework, I developed a framework for writing prompts in QDA called "ACTOR" (Table 2). This abbreviation stands for *Actor, Context, Task, Outputs,* and *Reference*, representing the five components in each prompt. *Actor* refers to the role-play prompting technique that assigns AI a specific role or persona (Kong et al. 2023). Outlining *tasks* in specific steps helps elicit chain-of-thought reasoning (Wei et al. 2022). Specifying *output* formats employs a template pattern technique (White et al. 2023), while providing examples of outputs or *contextual* information (e.g., examples of qualitative analysis concepts) can be categorized as a few-shot technique (Ma et al. 2023). *Reference* can serve either as the data source (inputs) for ChatGPT to analyze or as output examples (e.g., *create the table following the format in the uploaded file*). However, a prompt does *not always* need to include all five components. For instance, if researchers already included the designed role of ChatGPT in the first conversation (*Actor*), they could omit this component in follow-up prompts—up to the point where context window is exceeded (see Sect. 6: Discussion).

4 Results

In this section, I present the Template Analysis process through four steps of my GAITA approach.

4.1 Step 1: Data familiarization

In this step, researchers typically engage in repeated reading to become familiar with their dataset and form a preliminary mental map of data locations and relationships. The question addressed here is how ChatGPT can assist with this process.

To evaluate, I first conducted data familiarization with one transcript (Interview Number 1 with Barbara). Initially, I performed the analysis step manually. This step included reading the whole transcript and producing my summary of the transcript to help grasp the whole meaning of each transcript. I recorded this process to see how long it took me to finish these tasks. Subsequently, I used GPT-4 to help grasp the content of the transcript, and finally, I compared its summary with mine. I used the following prompts to ask ChatGPT to generate different summaries of the transcript.

Prompt 1.1 *Can you please provide a summary of this interview [Interview number 1 with Barbara]?*

Table 2 The ACTOR framework for prompting in qualitative data analysis

Components	Description	Example
Actor	Specifies the role of ChatGPT, associating it with specific functions, clearly defining its operational domain	"You're a qualitative research assistant"
Context	Includes essential background information about the task (e.g., research questions), explaining concepts (e.g., code, cluster, theme) and domain knowledge necessary for ChatGPT to understand and execute its tasks effectively	"Code is defined as…"
Task	Clearly defines the specific actions ChatGPT is expected to perform, such as summarizing transcripts, as well as actions it should avoid, such as producing overlapping codes	"Can you please *summarize* this transcript…?"
Outputs	Describes the expected end product in detail, providing clear examples of what the outputs should look like, including the format and structure	"…in three key bullet points"
Reference	Provides specific data or resources that the AI should reference or analyze during its task execution, such as transcripts, documents, or datasets, to ensure accurate and relevant outputs	"###Here is the transcript…"

Source: Adapted from Lin (2024), Eager (2023), Presseau et al. (2019)

Prompt 1.2 *Highlight the key ideas of the interview in three key bullet points.*

It took me, on average, 11 min to read over one transcript without taking any notes, as I usually did, and about 5–6 min to produce a 207-word summary without paying attention to grammatical and spelling errors. GPT-4 produced a summary of 419 words in about 30 s. Its summary includes 7 paragraphs, which consist of subheadings such as Personal History and Connection to Down East, Connection to Down East's Natural Environment, Professional Perspective on Down East, Community and Environmental Changes, and Vision for the Future of Down East. In doing so, GPT-4 did not repeat information already stated. When generating the three bullet-point highlights, GPT-4 finished in 9 s. GPT-4 displayed a high level of understanding of the transcript, highlighting the distinction between the culture-community-environment nexus, key environmental challenges, and Barbara's vision for the future (see Appendix 1).

Proceeding to a summary of the entire dataset, I took advantage of GPT-4's ability to scan and respond to queries about the files I uploaded to the chat window. I created a file named 'All 11 interviews' containing all 11 transcripts, with 20,832 words (including the headings of each interview I added). Here is the prompt for GTP-4.

Prompt 1.3 *Can you please provide me with a 400-word summary of key ideas from all the interviews in the uploaded file? All interviews are numbered, and each starts with a short introduction of the interviewees.*

GPT-4 generated a summary (see Appendix 2) in just 10 s. In this summary, ChatGPT was not only able to provide the summary of the key points but also identify key themes of the dataset (in its own definition of theme), including deep community roots, environmental connection and concerns, economic shift, cultural and social change, visions for the future, and local knowledge and wisdom, for all interviews. This suggests that GPT-4 has the capacity to generate themes once it has the raw data. However, as I suggested above, I avoided using its produced themes at this stage to allow me to have a meaningful engagement with the data.

4.2 Step 2: Preliminary coding

This step contained two tasks: one was to start preliminary coding of the first transcript to produce a list of initial codes, and the other was to use this list to code and recode the rest of the transcripts. For the sake of testing GPT-4, in this task I focused only on two questions out of six in the Down East's interview, including "Q1. Connection to Down East" and "Q2. Connection to Down East's natural environment." I also needed to provide GPT-4 with the research question "How do these people connect to Down East?" to set up the necessary context. From this step, I created a Google Sheets file to record and manage GPT-4's outputs.

4.2.1 Coding the first transcript

In this task, I gave ChatGPT the following prompt to work with the first transcript, namely, Barbara's interview. In Prompt 2.1, I incorporated the definition of codes from Saldaña (2016) to help train GPT-4.

Prompt 2.1 *You're a qualitative research assistant. You will help me identify the relevant codes from the following transcripts in response to this question: "How do*

these people connect to Down East?". Codes are labels that assign summative, salient, essence-capturing, and/or evocative attribute/ meaning to a portion of data. #The final outputs are a table like this:
Column 1: Code; Column 2: Description of code meaning; Column 3: Quotation representing the code from transcript.
##Here is the transcript [insert the transcript]

In its result, GPT-4 was able to produce five initial codes that match the datum (i.e., statements from the interviews). Despite this, GPT-4's labeling could be more meaningful. For instance, the in vivo code "feeling local" extracted from the phrase "feel kind of like a local," could replace the code "Connection as Non-Local." Based on my own coding of Barbara's interview (Appendix 3), I also saw that "natural beauty" was one of the key ideas in Barbara's connection to Down East. As such, I asked GPT-4 to revise the list based on my suggestion. Table 3 includes a revised list of six codes. The content in […] were my edits, while some strikethroughs were the deleted GPT-4-generated content based on my judgement.

4.2.2 Coding the rest of the transcripts

The next task was to apply the revised list of initial codes to the remaining transcripts (10 interviews) from the 10 remaining interview transcripts. I created a file named "10 interviews" containing 5,682 words, including answers to two questions, Q.1. Connection to the Down East and Q.2. Connection to the Down East natural environment. I then asked GPT-4 to study the revised list of initial codes and prepare for my next request (Prompt 2.2). GPT-4 responses showed that it was able to perform the task I requested. In my first attempts, I focused on testing GPT-4's capacity to code the whole file by asking it to analyze the entire dataset (20,832 words) (see Prompt 2X, Appendix 4). GPT-4's output was too brief. Each code only contained quotes from one interview, which risked losing much information from the other interviews. When I asked GPT-4 to produce a Google Sheets file to ensure the detailed level of coding, it acknowledged, "Given the volume of text and the depth of analysis required, compiling and summarizing this information into a Google Sheet file is a substantial task that requires significant time and attention to detail. Unfortunately, I cannot directly produce or share files." This result shows that GPT-4, despite its advanced technologies, was still unable to work with a large qualitative dataset at once. If we tried to ask it to do so, its coding would be superficial.

After this unsuccessful attempt, I worked with each transcript one by one. I broke the prompts into smaller tasks. In Prompt 2.2, I asked GPT-4 to get familiar with the current list of initial codes based on Barbara's interview, then in Prompt 2.3, I asked this tool to apply the list to code the remainder of the transcripts.

Prompt 2.2 *Please study the following list of initial codes developed based on the interview with Barbara. We will use this list to code relevant information in response to the question, "How do these people connect to Down East?" In the next question, I will ask you to apply this list in table format to code the rest of the dataset. ### Here is the list of codes [insert the revised list of initial codes based on Barbara's interview here].*

Prompt 2.3 *Now, apply the revised coding framework based on the interview with Barbara to the following interview in response to "How do these people connect to*

Table 3 A revised list of initial codes based on Barbara's transcript

Code	Description	Quotation from Transcript
1. Family ~~Connection~~ [history]	Connection to Down East through family history and upbringing	"My family moved here when I was two years old in 1969. My parents still live here. They live down in Gloucester. But I was raised in Beaufort…"
2. Duration of Residence	The length of time Barbara and her family have lived in Down East, both full time and part time	"Since'96. My husband and I bought this little cottage in'96… It was full time, but then in 2000 he got a job up North…"
3. Connection as Non-Local [Feeling local]	Despite living most of her life in Down East, Barbara doesn't feel she is considered a local, yet she feels connected as one	"Yeah. And I'm definitely not considered a local. Even though I feel kind of like a local."
4. Attraction to Community	The desire to raise a family in a strong community ~~and the beauty of the area~~ attracted Barbara to return to Down East	"At that time I thought I'd like to have kids and I wanted to raise them in a place that had this strong community that I enjoyed growing up…"
5. Unique Cultural Experience	Barbara's perception of what makes Down East unique, including the distinct communities and colourful language and culture	"I think it is just how distinct the different communities are because of the geographical isolation and that—all the different accents…"
6. Natural Beauty [added]	Appreciation for the area's natural landscapes and environment	"I've always thought it was some of the most beautiful beaches and just water anywhere." (Barbara)

*Down East?" # The final output must be a table with the following columns: code, description, and interview quotes, including the interviewee's name. ## When doing so, you should: (1) maintain the code order in the current coding framework; (2) edit the current coding framework and/or add new codes to the end of the table to reflect new information. When there is a new code, please indicate by adding *; (3) edit the coding description to reflect your change; (4) add quotes from the interview to illustrate the code. In case there are no relevant direct quotes, please summarize the information from the interview. ### Here is the interview with Charles [Charles' interview here].*

GPT-4's response to Prompt 2.3 was significantly improved. GPT-4 was able to add one more code, "7. Engagement with Local Industry," based on the new information from Charles' interview. When there was no information matching the code, it responded, "No direct quote." I continued to use Prompt 2.3 for the rest of the interviews. After coding one transcript, GPT-4 produced a table with a list of codes and corresponding quotations. In this step, I needed to verify if a single quotation was accurately reflected in the corresponding code. After coding the first five transcripts including Barbara's interview, the list of preliminary codes created by GPT-4 contained 10 codes —4 more than the initial list. This list was consistent, requiring no further revisions from me. From then on, GPT-4 maintained this number of codes and did not produce any new ones. To ensure completeness, I added this question, "Is there any new information that would help form new codes?" after each round of coding. GPT-4 would explain that there is no new information and that the current list of codes is already comprehensive. Table 4 presents the comparison between the list of initial codes based on Barbara's transcript and the refined list of codes after coding the whole dataset. This was the result of my revising and editing with the support of GPT-4. The next step is to apply this refined list to form clusters.

4.3 Step 3: Template refinement

The goal was twofold: (1) to condense data by building clusters or categories, which are more abstract than preliminary codes; and (2) to modify and finalize the initial template. Prompt 3.1 asked GPT-4 to group the initial codes in Step 2 (Table 4) into meaningful clusters.

Prompt 3.1 *From the above refined list of codes, please group codes into clusters (i.e., more abstract codes) in response to the question: "How do these people connect to Down East?".*
#For each of the clusters, please retain the specific codes and quotations from specific transcripts. If two or more transcripts fall under the same cluster, please put them together.
##The final output will be a table: Column 1: Cluster; Column 2: Codes; 3. Description of Cluster Meaning; Column 4: Quotation from Transcripts. Please make sure you quote the names of the interviewees for each quote.

GPT-4 produced a table with four clusters (Appendix 5) while retaining codes and representative quotes. However, GPT-4 omitted the initial code "7. Engagement with local industry" in this table. Therefore, I asked it to restore the code into the table and create a fifth cluster. I also added the word "family" to the first cluster to make it more meaningful. Table 5 presents the initial template in King et al.'s (2018) sense.

Table 4 Comparison between the initial and refined lists of codes

List of initial codes	Refined list of codes	Code description
1. Family history	1. Family history	Connection to Down East through family history and upbringing
2. Duration of Residence	2. Duration of Residence	The length of time spent living in Down East, both full time and part time
3. Feeling local	3. Feeling local	Feel as a local in Down East
4. Attraction to Community	4. Attraction to Community	The desire to raise a family in a strong community and the beauty of the area attracted Barbara to return to Down East
5. Unique Cultural Experience	5. Unique Cultural Experience	What makes Down East unique is its distinct community with colourful language and culture
6. Natural Beauty	6. Natural Beauty	Appreciation for the area's natural landscapes and environment
N/A	7. *Engagement with Local Industry	*Engagement in local fishing industry and integration into local economic activities
N/A	8. *Local Traditions and Activities	*Engagement in traditional local activities and appreciation for the lifestyle
N/A	9. *Advocacy for Local Industries	*Active leadership and involvement in initiatives to support and sustain local industries
N/A	10. *Environmental Awareness	*Strong awareness and concern for environmental issues affecting Down East

*Indicates the new codes after coding the remaining transcripts

Table 5 Initial template based on clustering

1. Family Connection to Place
1.1 Family history
1.2. Duration of Residence
2. Cultural Connection
2.1. Feeling loca
2.2. Unique Cultural Experience
3. Community Engagement
3.1. Attraction to Community
3.2. Local Traditions and Activities
4. Appreciation of Environment
4.1. Natural Beauty
4.2 Environmental Awareness
5. Advocacy for local industry
5.1 Engagement with Local Industry [re-added by the author]
5.2. Advocacy for Local Industries Economic Participation

With this initial template, the next task was to modify and ultimately finalize the template. This task would involve recoding and recategorizing data. First, to ensure capturing nuances in the data, I asked GPT-4 (Prompt 3.2) to create additional subcodes for each cluster based on the previously identified quotations from each transcript (see Prompt 2.3, Step 2).

Prompt 3.2 *Now I need you to further develop the coding table above by generating subcodes for each code based on the quotation I give you. I will provide you with the quotations under each code. #Task: You need to (1) add those quotations to the suitable subcodes, (2) create new subcodes to label these quotations' contents. The final output is a table with this structure: cluster, code, subcodes, description of subcodes, quotations. ###Here is the first set of quotations under "family history" [paste the data… and repeat with the code "Duration of residence"]*

Then I repeated Prompt 3.2 for the code "duration of residence." GPT-4 generated two tables of clusters, codes, and subcodes for each of the above codes. Table 6 contains the results generated by GPT-4 for the first cluster, "Family Connection to Place."

As shown in Table 6, the two codes "Family history" and "Duration of residence" have many overlapping subcodes (e.g., "generational residence" versus "settled for generations"). Although I asked GPT-4 to remove the redundancies, it still repeated some overlapping contents. In response, I intervened and made subjective decisions. Prompt 3.3 guided GPT-4 on how to arrange the codes and subcodes in a more meaningful way.

Prompt 3.3 *The two tables are still somewhat overlapping. Please create one new table combining two coding tables. In this table, please build the coding structure as follows: 1. Family history: 1.1. Generational ties (settled for generations); 1.2 Early Childhood or Upbringing; 2. Duration of Residence: 2.1 Acquisition of Property; 2.2. Long-term residence; 2.3 Recent relocation. Please also add any new subcodes or codes that you find missing from my suggested list.*

Table 6 GPT-4's subcode generation for the cluster "Family Connection to Place"

GPT-4's first recoding cluster "Family Connection to Place"
1. Family Connection to Place
1.1 Family history
1.1.1 Early childhood relocation
1.1.2 Generational residence
1.1.3 Long-term family presence
1.1.4 New generational roots
1.2 Duration of residence
1.2.1 Recent acquisition
1.2.2 Long-term residence
1.2.3 Settled for generations
1.2.4 Recent relocation

Table 7 Researcher and GPT-4's revised coding for Cluster "Family Connection to Place"

1. Cluster: Family Connection to Place
1.1. Family history
1.1.1. Generational Ties
1.1.2. Early Childhood or Upbringing
1.1.3. New Family Integration [new code]
2.1. Duration of Residence
2.1.1. Acquisition of Property
2.1.2. Long-term Residence
2.1.3. Recent Relocation
2.1.4. Seasonal Residence [new]
2.1.5. Return After Absence [new]

Responding to this prompt, GPT-4 suggested new subcodes. This revised list of codes and subcodes under Cluster "Family Connection to Place" is presented in Table 7.

In applying the same procedure (Prompts 3.2 and 3.3) to the remaining clusters, GPT-4 continued to produce overlapping codes and subcodes or to place them in inappropriate clusters. During this process, I used the Google Sheets file and edited the GPT-4 recoded list. This task required my own intellectual effort in judging and making decisions on how to better arrange codes, subcodes, and clusters. After ten rounds of recoding and recategorizing, I arrived at the final template with relatively distinctive clusters and codes (Table 8).

4.4 Step 4: Theme development

In this step, I asked GPT-4 (**Prompt 4.1**) to develop a theme table based on the final template. The definition of a theme was taken from King (2012, pp.431–432).

> **Prompt 4.1** *From the above table of clusters, please generate themes across the clusters and codes in response to the question: "How do these people connect to Down East?". #Theme is defined as the recurrent and distinctive features of par-*

Table 8 Final template

Cluster	Code	Subcode
1. Family connection to Place	1.1 Family history	1.1.1 Generational Ties
		1.1.2. Early Childhood or Upbringing
		1.1.3. New Family Integration
	1.2 Duration of Residence	1.2.1. Acquisition of Property
		1.2.2. Long-term Residence
		1.2.3. Recent Relocation
		1.2.4. Seasonal Residence
		1.2.5. Return After Absence
2. Cultural Connection	2.1 Feeling Local	2.1.1 Self-Perception as Local
		2.1.2 Community Recognition
		2.1.3 Self-Roots
	2.2 Unique Cultural Experience	2.2.1 Linguistic Uniqueness
		2.2.2 Local Culture and Heritage
		2.2.3 Architectural uniqueness
		2.2.4 Talent and Artistry
3. Attachment to community	3.1. Engagement in Local Activities	3.1.1. Swimming and Water Activities
		3.1.2 Clamming
		3.1.3 Fishing and Shelling
		3.1.4 Seafood Gathering
		3.1.5 Seasonal Visits
		3.1.6 Flounder Gigging
		3.17 Boating and Exploration
	3.2 Community Belonging	3.2.1 A big family
		3.2.1.1 Inherited Land Connection
		3.2.1.2 Familiarity and Friendship
		3.2.2 Peace in Belonging
		3.2.3 Community Support
	3.3 People	3.3.1 Real People
		3.3.2 Hard working people
		3.3.3 Commercial fishermen
	3.4 Desirable Community Values	3.4.1 "Strong community"
		3.4.2 "Raising children"

Table 8 (continued)

Cluster	Code	Subcode
4. Appreciation of Environment	4.1 Water Connection	4.1.1 Beaches and Waterways
		4.1.2 "Clean water"
		4.1.3 Personal History with Water
	4.2 Natural Beauty	4.2.1 Marshland Connection
		4.2.2 Core Sound's Beauty
		4.2.3 "Magical Place"
	4.3 Desire for Preservation	4.3.1 Desire for Natural Preservation
		4.3.2 Beauty of Remoteness
		4.3.3 "Rural feel"
	4.4 Environmental vulnerability	4.4.1 Fragility of Environment
		4.4.2 Development Concerns
		4.4.3 Concern for water quality
		4.4.4 Protecting Environment
5. Industry Involvement	5.1. Engagement with Local Industry	5.1.1 Direct Involvement
	5.2 Advocacy for Local Industries	5.2.1 Organizational Leadership

Note: I have excluded quotations for each subcode from this table

ticipants' accounts that characterize perceptions and/or experiences, seen by you as the researcher as relevant to the research question of a particular study. Each theme should link the cluster(s) and/or code(s) with the context of Down East. Themes must be relatively distinct from each other, although some overlap is inevitable. ##The final output will be a table: Column 1: Theme; Column 2: Clusters and Codes used for the theme; 3. Description of Theme Meaning; Column 4: Quotation from Transcripts. Please make sure you quote the names of the interviewees for each quote.

GPT-4 generated a table with six themes, including Generational Anchoring; Long-Term Residence and Attachment; Local Identity and Cultural Pride; Community Engagement and Belonging; Environmental Connection and Stewardship; Economic and Industry Advocacy. This result shows that GPT-4 is capable of producing themes across different clusters and codes. Yet, the suggested themes do not appear significantly different from clusters, although I had provided the definition of theme in Prompt 4.1. This might be because that GPT-4 did not 'understand' the concept of "theme" in qualitative research. Themes are different from categories or clusters because their meanings are well situated in the socio-economic contexts of the research sites (Saldaña 2016). To make themes more meaningful, it is necessary to guide GPT-4 to extend the themes' names that so that they are connected back to the context and our research question (see Saldaña 2016). Thus, I asked GPT-4 to revise the list of themes (Prompt 4.2).

Table 9 Revised list of themes

1. [Connection through] Generational and Historical Continuity: Individuals' connections to Down East are often anchored in family history, with deep generational roots or new family integration into the community, emphasizing a continuity that spans decades or even centuries
2. [Connection through] Cultural Identity and Local Pride: There is a strong sense of local identity, with people feeling [proud of] the unique cultural fabric of Down East, characterized by [genuine people], distinctive language, traditions, values, [and a sense of belonging and support within the community structure] that are proudly maintained and celebrated
3. [Connection through] Active Community Engagement: The theme highlights active participation in and contribution to community life, through engagement in local activities and traditions, [maintaining relationships with local people]
4. [Connection through] Appreciation and Advocacy [for Natural Beauty and Environmental Sustainability]: A significant aspect of the connection to Down East is the reverence for its natural beauty and environment, coupled with a commitment to preserving its pristine quality and addressing environmental vulnerabilities
5. [Connection through] Local Livelihood and Lifestyle: This theme represents Down East as a place where economic activities, particularly fishing, and traditional practices form a core part of life, offering both livelihood and meaning

Prompt 4.2 *Can you revise the themes' names and descriptions by: (1) extending the theme names so that they are meaningfully connected to the social context of Down East; (2) ensuring that the themes and their descriptions answer the question "How do these people connect to Down East?*

Based on my suggestion, GPT-4 was able to reproduce the following themes (note: words in […] are my addition/edits (Table 9). I added the words "connection through" to make the themes more meaningful and connected to my research question.

After having the five themes, I asked GPT-4 to produce an overarching theme.

Prompt 4.3 *From the themes above, please produce an overarching theme in response to the question "How do these people connect to Down East?"*

Table 10 shows that GPT-4 was able to connect five themes in response to the question "How do these people connect to Down East?". Its selection of the central theme "Rootedness and Reverence in Down East" effectively captures the multiple connections and sense of belonging that research participants felt towards Down East. Yet, to better align with the research question, this label could be further refined by adding the phrase "Connection through" to make it clearer.

Table 10 GPT-4's generation of an overarching theme

The overarching theme in response to "How do these people connect to Down East?" is "[Connection through] Rootedness and Reverence in Down East" encapsulates the profound and multifaceted connections that individuals have with the region. This theme reflects not only a deep-seated sense of belonging and attachment stemming from generational ties and long-term residence, but also a profound respect and appreciation for the unique cultural, environmental, and economic aspects of Down East

5 Discussion

Through the application of ChatGPT across nearly the full circle of Template Analysis—a method that remains relatively underexplored in the current experiments with GenAI—my study contributes to ongoing scholarly attempts to test the capacity of GenAI to analyze qualitative data. My study provides a detailed how-to guide termed Guided AI Thematic Analysis (GAITA), which adapts King's (2018) Template Analysis into a four-step process involving the use of GenAI. In response to the call for prompts designs tailored to qualitative data analysis (QDA) purposes (De Paoli 2024), I propose the ACTOR framework (*Actor, Context, Task, Output,* and *Reference*) that supports researchers in crafting effective prompts using techniques such as chain-of-thought, template, few-shot, and role-play. Furthermore, the GAITA framework emphasizes the interactive nature of qualitative analysis, highlighting the importance of researcher knowledge, judgement, and interpretation in guiding GenAI in the analysis process. Throughout this experiment, I advocate for a human-AI collaborative approach, where researchers retain their decision-making and leadership in monitoring, evaluating, critically reflecting on and learning from GenAI as an assistant in QDA.

The use of ChatGPT in QDA presents several methodological implications that need thorough consideration. Below I discuss some potential benefits and limitations based on my experiment and offer some methodological and practical recommendations for working with GenAI in QDA.

5.1 Understanding GenAI's potentials

5.1.1 GenAI enables efficient code generation and data display

GenAI can assist in generating initial codes and arranging coding hierarchies, which can be a good starting point for more in-depth analysis. In my study, GPT-4 was able to code the first transcript of Barbara's interview and produce an initial list of codes. Subsequently, GPT-4 could apply this coding framework to the remaining interview transcripts, and even identify new codes when new information arose. However, GPT-4 sometimes produced overlapping codes and misclassified similar codes under different clusters. This situation required me to carefully evaluate GPT-4's outputs and make decisions about (re)naming, (re)arranging, and (re)allocating the codes and clusters. Researchers' intervention is thus critical in modifying the template and developing the final template. As with manual coding, we must guide GPT-4 through multiple rounds of coding and recoding to ensure it captures the nuances from interviews. Since GPT-4 lacks specialized knowledge of QDA, researchers need to explicitly direct it towards specific types of codes, such as an "in vivo" code or a "process" code (De Paoli 2024), as it tends to default to descriptive codes—*topic-based nouns* (Saldaña 2016).

GenAI is also helpful in facilitating *data display*—an "organized, compressed assembly of information that permits conclusion drawing and action" (Miles and Huberman 1994, p. 11). Particularly, ChatGPT could turn unstructured text into more organized through diagrams, flowcharts, networks, or matrices that arrange data segments, codes, memos, and themes according to our reasoning and interpretation.

5.1.2 GenAI can assist in pattern recognition and theme generation

Beyond coding, my findings show that GenAI demonstrates an ability to work across different codes and clusters to produce (overarching) themes, moving toward more abstract levels of analysis. Based on its diverse training data, ChatGPT can grasp basic concepts such as community, environmental stewardship, and culture. Through sustained interaction with researchers, GenAI can be trained to work with more complex concepts deriving from research questions and qualitative research methodologies such as code, cluster, and theme. In this regard, ChatGPT or GenAI can be helpful in recognizing patterns of meaning across the dataset and generating themes in response to our research questions.

While I agree with Morgan's (2023) observation that GenAI often struggles with constructing interpretative themes, my findings suggest that GPT-4, when provided with proper guidance throughout a full analysis process, can meaningfully connect different data layers (e.g., clusters, codes, subcodes, data segments) to assist us in constructing more contextual and interpretative themes. My current study is limited in fully executing Template Analysis' final steps—namely, using the final template for interpretation and report writing. Future research should close this gap, thereby evaluating GenAI's capacity to produce deeper interpretative insights.

5.2 Recognizing GenAI's limitations

Despite its potential, the current ChatGPT model has some key limitations for QDA. I offer some practical solutions to address them.

5.2.1 GenAI has processing constraints with large datasets

While GPT-4 improves upon GPT-3.5 through its file upload function, allowing users to input extended text and multimodal files directly into conversation, its capacity to do complex tasks involving long-text files and multiple steps remains limited. This limitation stems from GPT-4's *context window*, defined as "the maximum number of tokens that can be used in a single request. This max tokens number includes input, output, and reasoning tokens" (OpenAI 2024). GPT-4's context window of 8192 tokens means that it can process up to approximately 6,144 words, considering inputs, outputs, and reasoning tokens (with each token equivalent to around 0.75 words in English). Given that my whole dataset (about 20,000 words) exceeded its capacity, GPT-4 could not thoroughly code the entire text corpus as instructed.

This technical constraint creates significant risks: when asked to analyze documents beyond its context window, GPT-4 (and GenAI) may oversimplify complex qualitative data as it can "forget" previous information, including established rules and data we gave them from earlier conversations. Rather than condensing the data (i.e., abstracting data while preserving its essential meanings) (Saldaña 2016), it tends to reduce data (i.e., omits portions) and could even hallucinate or make up responses (Lakshmanan 2022, Morgan 2023). Consequently, researchers must understand each AI tool's context window limitations to design appropriate prompts and manageable datasets.[2] One practical solution is to divide the dataset into smaller segments or chunks of text and process them sequentially, which facilitates closer researcher supervision of AI-produced outputs.

[2] Currently, according to the report from Artificial Analysis (2024), Google's Gemini 1.5 Pro has the largest context window of 2 million token, followed by Jamba 1.5 Large (256,000 tokens), Anthropic's Claude 3.5 Sonnet (200,000 tokens), OpenAI's GPT-4o and o1-preview (both 128,000 tokens).

5.2.2 GenAI tends to produce variable and inconsistent outputs

Although it is easy to communicate with ChatGPT, as Morgan (2023) suggests, getting what we want through ChatGPT is no straightforward task. The prompts and steps that I presented above were not the result of one but many attempts to correct the prompts or request ChatGPT to repeat its results. For instance, sometimes GPT-4 did not follow my prompts to create a table with desired columns; instead, it only presented a quick summary. In this case, I had to repeat or adjust the prompts several times before obtaining the results in the desired format.

5.2.3 Using GenAI chatbots requires working across multiple platforms

One of the advantages of CAQDAS programs is that they create a single workspace where researchers can link everything from data to codes, from codes to clusters and themes to notes, and can retrieve them anytime. While GPT-4 did not offer the capacity to retrieve contents across its conversation, current ChatGPT or Claude models offer this funtionality. The larger issue, however, is that we need to navigate through different windows and programs when using a GenAI chatbot. In my case, I needed to arrange the original dataset in Microsoft Word, uploaded files into ChatGPT, then exported the results back to Microsoft Word or Google Sheets for further work (linking, categorizing, and memo-writing). While GenAI chatbots currently allow users to import data from Google Drive or a local computer and export files into specific formats (e.g., Rich Text Format, Word, Excel) and allow to organize conversations into folders (e.g., the feature of "Projects" in Claude), users still need to work across multiple platforms. Consolidating the analysis process and results into a single Google Sheet file is helpful in this case.

5.3 Developing a systematic approach to ensure methodological rigor

To revisit Gerson's (1984) hypothesis that AI should possess "sociological knowledge" to serve as a research assistant, we can argue that GenAI now possesses general knowledge of many topics because of its vast amount of training data, including its capacity to understand common concepts such as family, traditions, environmental stewardship, and so forth. Yet, GenAI in general and ChatGPT in particular, is not trained in the proper knowledge and rules required for qualitative research. For instance, in my study, GPT-4 generated themes that were indistinguishable from clusters, and as a result, initial themes lacked contextual meaning connected to the research participants.

To address this limitation, we need to train GenAI in QDA principles and concepts (De Paoli 2024). This might involve some technical configuration within the GenAI system. For instance, in ChatGPT, we can design "Custom Instructions" (this is also available for the free version), which is a set of rules and principles that tell ChatGPT what it should and should not do when handling qualitative data. For instance, we need to tell it not to make up any data and strictly use the data from the transcripts we uploaded. For ChatGPT's paid version, users can now create a *customized GPT* for QDA purposes (Claude offers similar functionality with its "Projects" feature), which allows uploading documents to its knowledge base. In these documents, we can include key concepts and principles in QDA that act as a guideline for GenAI's data processing and responses.

More importantly, qualitative researchers need a systematic approach to guide and train GenAI in establishing trustworthiness in qualitative data analysis. This approach should highlight the primary role of the researchers in engaging with the data, meaning-making, analyzing data, and making final decisions about the interpretation (Paulus and Marone 2024). This approach should also enable researchers to monitor, evaluate, self-reflect, and learn from their interaction with GenAI. Furthermore, qualitative researchers as the "instrument of study" (Watt 2007) must be responsible for the knowledge they produce in engaging with data.

Additionally, to ensure methodological rigor, our approach needs to develop various means such as triangulation, documentation, researchers' reflexivity, and audit trails (Nowell et al. 2017). In this paper, to enhance methodological rigor, I offer a detailed four-step Template Analysis framework or a Guided AI Thematic Analysis (GAITA) with clearly defined prompts, AI's tasks, and researchers' roles. In particular, the GAITA approach offers a *reflexive space* for researchers to think deeply about *their* inputs (e.g., prompts, tasks, expectations of AI), evaluate AI-generated outputs, and document their decision-making process in each analytical step. This space enables an ongoing, iterative process for monitoring, evaluation, self-reflection and learning throughout the analysis process. This systematic approach allows for AI integration in thematic analysis while maintaining "trustworthiness" as proposed by Nowell et al. (2017). Therefore, instead of developing a procedure that lets GenAI work as *autonomous agents*, in this paper, I champion the process that researchers take GenAI as their assistant and maintain the lead throughout the analytical process. GenAI can support here and there, but in the end, researchers must be able to claim that they produce the knowledge through their intellect, positionality, and reflexivity.

6 Conclusion and implications

"[D]umping my text into a program and seeing what comes out" (Weitzman 2003, p. 315) might not be possible with most of CAQDAS (except Atlas.ti, MaxQDA and NVivo with AI functions), but it is now possible with GenAI chatbots such as ChatGPT, Claude, Copilot, or Gemini. However, this comes with the risks of inappropriate uses of GenAI for qualitative research's purposes. Instead, researchers working with these AI models should *always* be careful with AI-generated outputs since they could hallucinate their responses and carry the biases from their trained data into their judgement and outputs (Yan et al. 2023; Parker et al. 2023). Additionally, the hesitation of many qualitative researchers in using AI and ML in analyzing qualitative data is understandable as the use of these advanced technologies seems to conflict with key principles in qualitative research (Chen et al. 2018; Paulus and Marone 2024). These principles include seeing QDA as an "art" rather than a science, which is driven by researchers' judgement, reflexivity, creativity, experience, worldviews, cultural beliefs, intuitions, non-linear thinking, and even biases (Braun and Clarke 2019, 2022; Woods, Macklin, et al. 2016). The use of researchers as a primary tool in building rapport with the research participants, experiencing the world firsthand, reflexively making sense of and interpreting the world in which the participants live, makes their analyses unique. Consequently, the researcher's role should not be replaced by machines with mechanical procedures (Woods, Macklin, et al. 2016).

I believe that this statement still holds, even with the recent advancements of AI-powered tools like ChatGPT. Although we can dump our data into ChatGPT and let it generate codes, clusters, or themes, it risks oversimplifying nuanced data, and more importantly, losing our identity as qualitative researchers. The warning from Lee and Fielding

(1991, p. 8) that "the untutored use of analysis programs can certainly produce banal, unedifying, and off-target analysis" is still valuable in the case of GenAI. I argue that despite the advancements in GenAI and potential future developments, it remains crucial for qualitative researchers and practitioners to be well-versed in QDA methodological theories. It is critical not to let AI tools dictate our analytical approaches; rather, we should always guide AI's behaviors with our own reflexivity, intellectual judgement, and expertise (Woods and Wickham 2006). In the end, what truly matters (and therefore makes *our* qualitative research unique) is the social construction of knowledge based on the interaction between the researcher and the participants.

7 Appendix

7.1 Appendix 1: GPT-4's summary of Barbara's transcript in three bullet points

Role	Summary of key ideas	Duration/words
GPT-4	- Barbara emphasizes the importance of Down East's unique culture, community, and environment, and notes the challenges in obtaining septic permits for local development due to low land levels and high water tables - She observes the decline in commercial fishing due to environmental degradation and market factors, identifies stormwater runoff as the primary culprit for water quality issues, and highlights the negative impact of excessive land clearing on local wildlife - Barbara's vision for Down East's future includes maintained water quality, environmental protection, sustainable fishing practices, and increased reliance on renewable resources. However, she sees resistance to zoning in the area as a significant hurdle to comprehensive planning and sustainable development	9 s 112 words

7.2 Appendix 2: GPT-4's summary of all interviews

The interviews in your file explore the personal, environmental, and professional perspectives of various individuals connected to the Down East region in North Carolina. Key themes from these narratives include:
Deep Community Roots: Many interviewees have long-standing family connections to the area, often spanning generations. This deep connection fosters a strong sense of community and belonging
Environmental Connection and Concerns: The natural environment of Down East, particularly its waterways and marshes, is a central aspect of life. Interviewees express concerns about environmental changes and challenges, including water quality and the impact of development
Economic Shifts: There's a recurring theme of transition from traditional industries, like fishing, to other forms of livelihood. This shift is often linked to environmental, market, and regulatory changes
Cultural and Social Changes: The interviews reveal tensions between maintaining traditional ways of life and adapting to new circumstances, including the influx of new residents and shifting economic realities
Visions for the Future: Despite varied backgrounds, many share a vision for Down East that balances economic development with environmental preservation and community values
Local Knowledge and Wisdom: The narratives underscore the importance of local knowledge in understanding and addressing regional challenges, particularly in environmental and community development matters
These interviews provide a rich, multifaceted view of life in Down East, reflecting its complexities and the diverse perspectives of its residents

7.3 Appendix 3: Code list of Barbara's transcript by the author and GPT-4

Author's coding	GPT-4's coding
1. Family history	1. Family Connection
2. Childhood and education	2. Duration of Residence
3. Property ownership	3. Connection as Non-Local
4. Living duration	4. Attraction to Community
5. Job	5. Unique Cultural Experience
6. Feeling local	
7. Natural beauty	
8. Good for children	
9. Strong community	
10. Distinct culture	

7.4 Appendix 4: GPT-4's coding of the whole dataset

Prompt 2X Please apply the coding framework above to coding each of the 10 interviews in the uploaded file in response to the "How do these people connect to Down East?". #You should code one interview at a time. #The final output is a table with columns: Code, Coding Description, Interview Quotes including interviewees' names. ##When doing so, you should: (1) edit the current codes and/or add new codes to reflect new information from new interviews; (2) edit the coding description to reflect your change; (3) add quotes from all relevant interviews and the interview ID

Example of GPT-4's responses for one code:

- Code: Family history
- Code description: Connection to Down East through family lineage and upbringing
- Quotation: "My family has been here since the 1700s…" (Interview 2)

7.5 Appendix 5: GPT-4's generation of the clusters

Cluster	Codes	Description of Cluster Meaning	Quotation from Transcripts
Connection to Place	1. Family history, 2. Duration of Residence	This cluster reflects the deep-rooted connection to Down East through family history and duration of residence	"We've been here at least two to three hundred years." (Susan); "I've owned my house in Straits Township, in Otway, for three years." (William)

Cluster	Codes	Description of Cluster Meaning	Quotation from Transcripts
Community Engagement	4. Attraction to Community, 8. Local Traditions and Activities, 9. Advocacy for Local Industries	Captures the engagement with and attraction to the local community, participation in traditions, and efforts to advocate for local industries	"The people here are friendly, and God-fearing…" (Patricia); "During the summer season, I am typically on my boat or my dad's boat…" (William); "And sometimes it requires an outside hand to help; and if I can be a part of that, then I will." (William)
Appreciation of Environment	6. Natural Beauty, 10. Environmental Awareness	Highlights the appreciation for the natural beauty of Down East and the awareness of environmental issues	"I'm very fond of it. I would hate to see it change significantly." (Robert); "I enjoy the fact that for the most part, the water, the bay is very clean." (William)
Cultural Connection	3. Feeling local, 5. Unique Cultural Experience	Focuses on the feeling of being local and the unique cultural experiences of Down East	"I can't imagine why anyone would want to leave to tell you the truth." (Thomas); "The brogue there, like in all of Down East is still intact, at least in part." (Susan)
Industry Involvement (omitted in the first attempt)	7. Engagement with Local Industry	Economic Participation	Engagement in local industries and economic activities

7.6 Appendix 6: Comparing three scenarios in the integration of AIs in QDA

Aspects	CAQDAS with GenAI function	New Qual AI	AI chatbots
Advantages	- Seamless integration with existing workflows - Leverages existing CAQDAS features - Lower learning curve - Single workspace	- Tailored for specific GenAI functionalities - May offer more advanced GenAI features - Single workspace - Potential for greater innovation	- Real-time assistance - Versatile in handling various tasks - Can be fine-tuned for specific research needs
Disadvantages	- Limited by existing CAQDAS architecture - May not fully utilise AI capabilities - Potential compatibility issues - Expensive	- Requires learning a new system - May lack some CAQDAS features - Expensive	- Limited to text-based interactions - May require extensive fine-tuning - Ethical considerations for data privacy - Less expensive compared to other options

Source: author

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Data availability The raw dataset used in this paper is publically available on Lumivero's website; see Lumivero (2023).

Declarations

Conflict of interest The author has no relevant financial or non-financial interests to disclose.

Ethical approval This study only used secondary data from the project Environmental in Down East includes 11 short interviews conducted in 2008–2009 with the permission by Lumivero (2023).

Declaration of generative AI and AI-assisted technologies in the writing process During the preparation of this work the author used ChatGPT in order to support the template analysis. Key prompts and results are presented in the paper. The author also used ChatGPT to check for grammatical and spelling errors with the prompt "Proofread this part, focusing on grammatical and spelling errors. Bold the words you changed". After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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